LEA LABS · APPLICATION DEVELOPMENT

Android app development, made understandable.

A visual first step into Android Studio, Kotlin, and useful mobile apps.

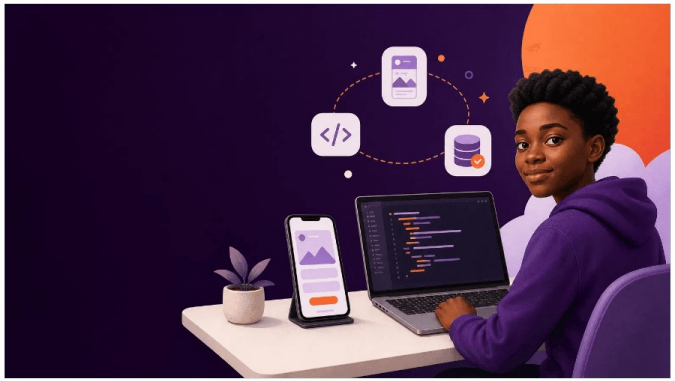
Beginner promise: start with a problem, build one small thing, and learn by testing it.

Start with one useful problem.

A good first app is not huge. It helps one person complete one job more easily.

Our class project: **Campus Tasks** — capture a task, mark it done, and find it later.

Ask before you code: “Who needs this?”



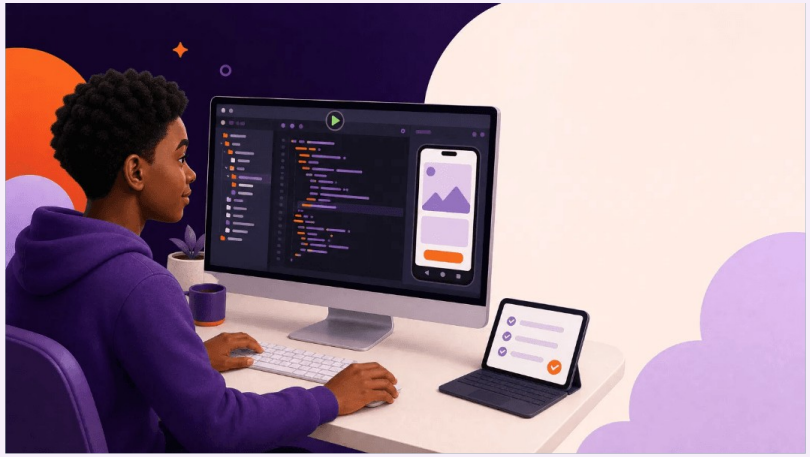
Screen
What a person sees and touches.

Logic
The rules that decide what happens.

Data
What the app remembers.

Quick prompt: What is one daily student problem a phone could reduce?

Android Studio is your workshop.



- Create the project, write code, preview the app, and fix mistakes in one place.
- The **emulator** is a practice phone inside your computer.
- Press **Run** to send your app to the emulator.

> **TRY IT**
>
> Find the project panel, editor, Run button, emulator, and Logcat before writing code.

Kotlin gives precise instructions.

Variable

Remembers a value, such as a task title or completion status.

Function

Groups one action, such as add a task or clear the input.

Class

Describes a useful thing, such as a `Task`.

```
data class Task(
    val title: String,
    val done: Boolean = false
)
```

Read it aloud:

“A task has a title and begins unfinished.”

Compose turns code into a screen.

- A **composable** is a reusable piece of interface: title, button, card, or task row.
- Describe what you want to see; Compose draws it for you.
- Keep each composable focused on one small job.

Try it: Change the sample title, run the app, and explain what changed.

KOTLIN CODE

```
@Composable
fun TaskRow(task: Task) {
Text(text = task.title)
}
```

↓

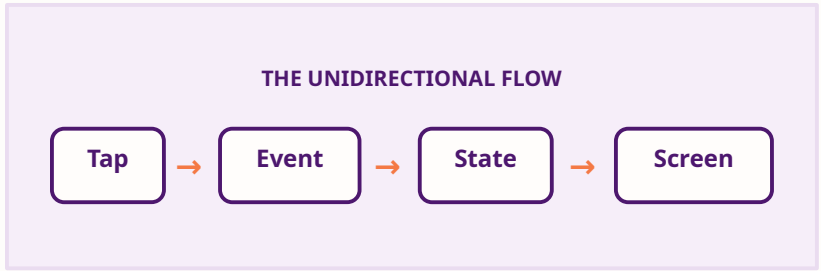
SCREEN OUTPUT

Buy groceries

State explains screen changes.

- **State** is the app's current memory: input text, selected item, task list, or loading status.
- A tap creates an event; the event changes state; the screen redraws.
- Keep one clear source of truth for each important piece of state.

```
var taskName by remember {
mutableStateOf("")
}
```



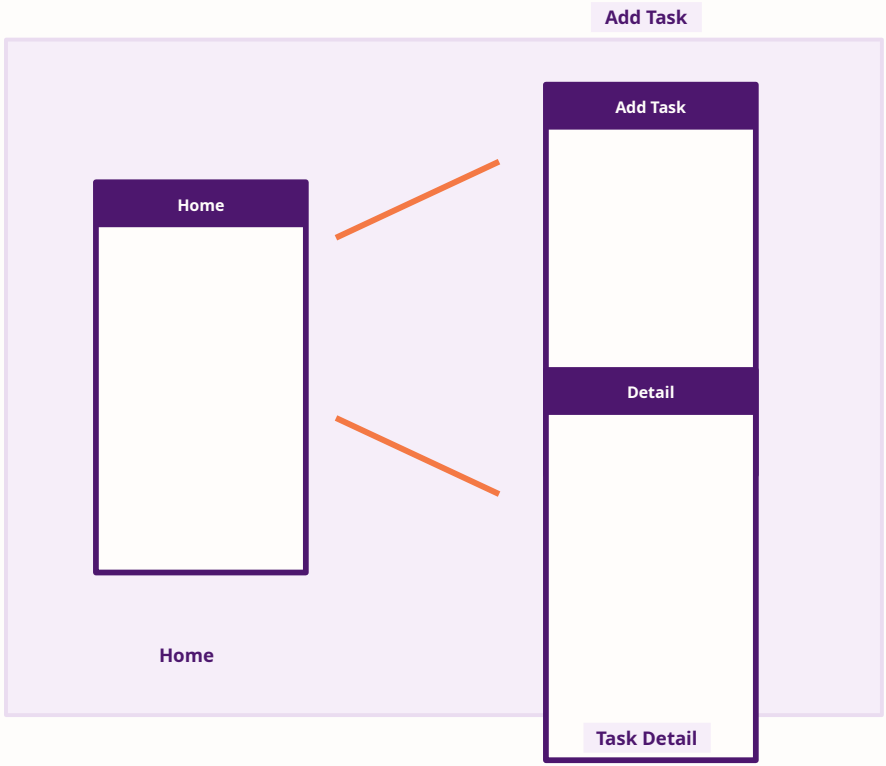
Pair activity: Act out the loop: tap → event → state → screen.

A small screen map prevents confusion.

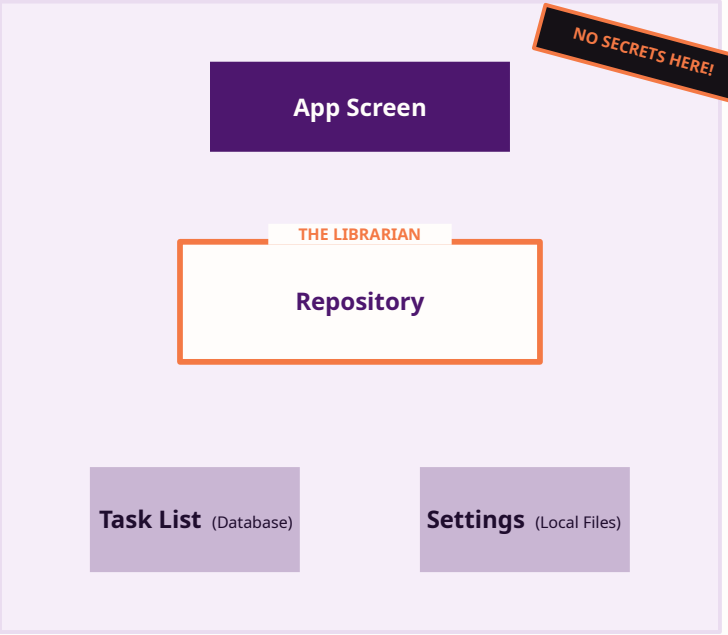
- Sketch the journey before coding: **Home → Add task → Task detail**.
- Give every screen one purpose and one obvious next action.
- Always provide a clear route back.

Design question:

What information must travel from Home to Task detail?



Good apps remember safely.



- Save simple settings locally; use structured storage for a growing task list.
- A **repository** gives the app one dependable place to request and update data.
- Never place passwords, API keys, or private secrets in the app code.

Memory aid:

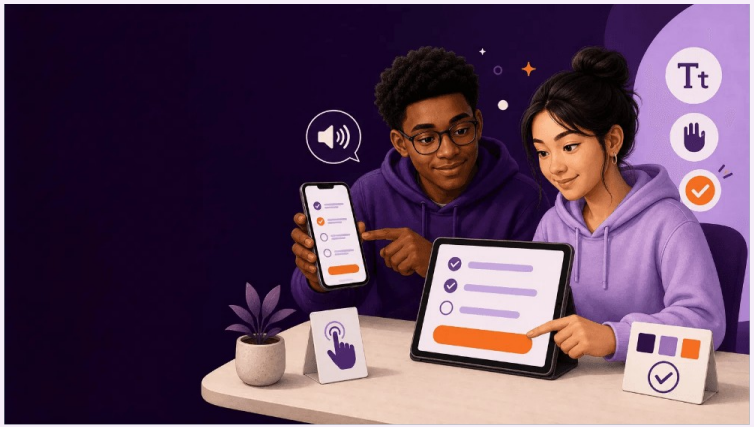
A repository is a librarian: it knows where the information belongs.

Build for more than one kind of user.

- Use clear labels, readable contrast, and buttons that are easy to touch.
- Let text grow without breaking the layout.
- Test important actions with accessibility tools and real people.

Accessibility check:

Can someone understand the task button without seeing its colour?



Test before you trust it.

01

Small Rules

Test the small rules first: *“Does a new task begin unfinished?”*

02

Full Journeys

Test the full journey on an emulator or a real phone.

03

Failure Cases

Test failures too: blank input, no connection, slow loading, and rotation.

Team challenge:

Write one failure case your classmates should try today.

Build a small app, then improve it.

1

WEEK 1

Plan the learner, problem, and three screens.

2

WEEK 2

Build a working screen, button, and simple task row.

3

WEEK 3

Add state, a list, and navigation.

4

WEEK 4

Save data, test, improve accessibility, and present your work.

FINISH LINE

A useful Campus Tasks app another student can actually test.

Keep learning with trusted guides.

OFFICIAL LEARNING

Android Basics with Compose

Google’s official beginner curriculum.

`developer.android.com/courses`

Jetpack Compose Docs

The complete guide to building UI.

`developer.android.com/develop/ui/compose`

ACCESSIBILITY & TESTING

Android Accessibility

Rules for building inclusive applications.

`developer.android.com/guide/topics/ui/accessibility`

App Architecture

How to separate screens from data safely.

`developer.android.com/topic/architecture`

YOUR NEXT STEP

Open Android Studio and start your first project.